



ANDERSON SERANGOON JUNIOR COLLEGE

2021 PRELIMINARY EXAMINATION

NAME: \_\_\_\_\_ (   )

CLASS: 21 / \_\_\_\_

**CHEMISTRY**

**8873/02**

Paper 2 Structured Questions

**14 September 2021**

Candidates answer on the Question Paper.

**2 hours**

Additional Materials:      Data Booklet

**READ THESE INSTRUCTIONS FIRST**

Write your name, class and register number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

The number of marks is given in brackets [   ] at the end of each question or part question.

| For Examiner's Use |   |      |                                               |      |
|--------------------|---|------|-----------------------------------------------|------|
| Section A          | 1 | / 14 | Paper 2                                       | / 80 |
|                    | 2 | / 10 | Paper 1                                       | / 30 |
|                    | 3 | / 5  | Percentage Overall                            |      |
|                    | 4 | / 10 | Grade                                         |      |
|                    | 5 | / 16 |                                               |      |
|                    | 6 | / 5  |                                               |      |
| Section B          | 7 | / 20 | Circle question number attempted in Section B |      |
|                    | 8 | / 20 |                                               |      |

This document consists of **25** printed pages and **3** blank pages.

## Section A

Answer **all** the questions in this section in the spaces provided.

- 1 (a) Fig. 1.1 shows the first ionisation energies of calcium to rubidium.

Use the *Data Booklet* and your knowledge of the variation in first ionisation energy of the elements of Period 3 (Na to Ar) to estimate and plot, on Fig. 1.1, the first ionisation energies of the elements: Ga, As and Kr.

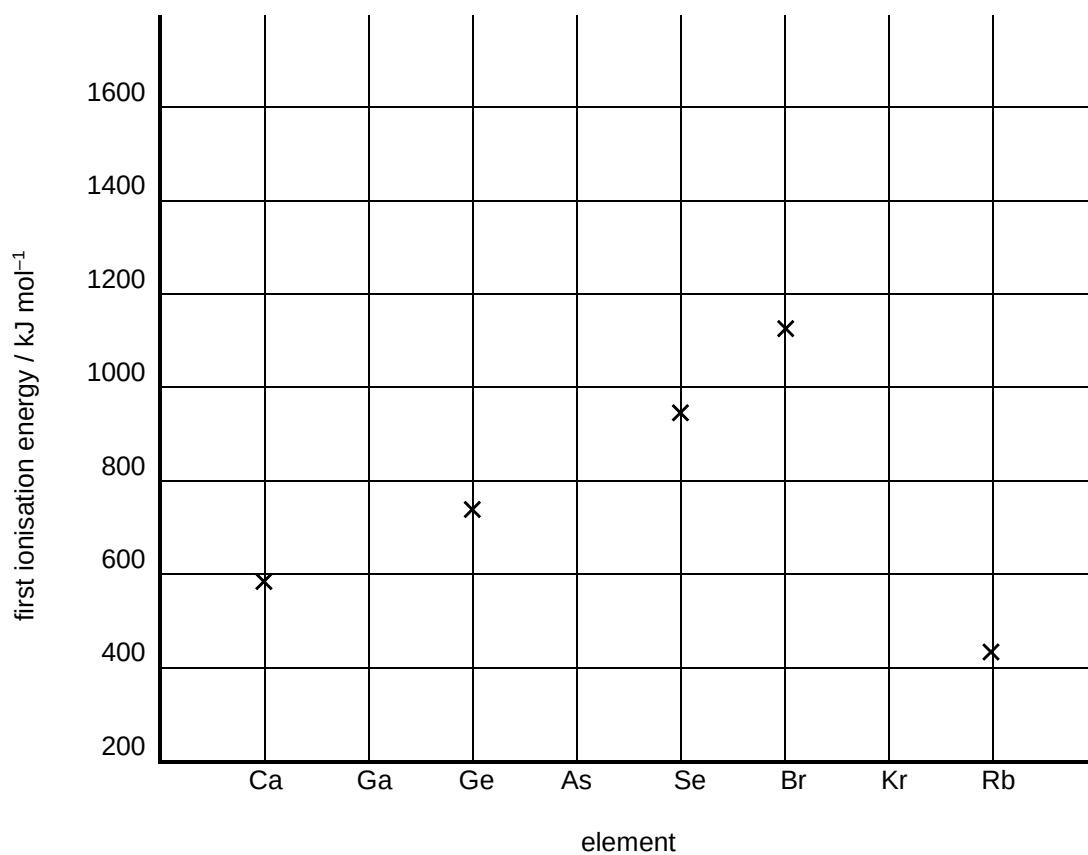


Fig. 1.1

[2]

- (b) (i)** Justify your prediction in **(a)**, by explaining why the first ionisation energy of As you have plotted is higher/lower/similar to Ge and Se.

[3]

- (ii)** Explain why the first ionisation energy of Kr you have plotted is higher/lower/similar to Rb.

.....[1]

- (c)** Draw the shape of the highest occupied orbital in Ca and Ga atom.

| Ca | Ga |
|----|----|
|    |    |

---

[2]

- (d) A student conducted some experiments involving Period 3 oxides and recorded his observations.

| element  | observations                                                                                                                            |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>W</b> | Oxide of <b>W</b> is a gas which dissolves completely in water to form a solution which turns blue litmus paper red.                    |
| <b>X</b> | Oxide of <b>X</b> is a white solid which dissolves completely in water to form a dark blue solution in presence of Universal Indicator. |
| <b>Y</b> | Oxide of <b>Y</b> does not dissolve in water and but reacts with concentrated NaOH on heating.                                          |
| <b>Z</b> | Oxide of <b>Z</b> can react with both aqueous NaOH and dilute HCl.                                                                      |

- (i) Identify the unknown elements.

| Unknown element | Identity |
|-----------------|----------|
| <b>W</b>        |          |
| <b>X</b>        |          |
| <b>Y</b>        |          |
| <b>Z</b>        |          |

[2]

- (ii) Oxide of **W** and oxide of **X** are mixed together in the same beaker of water. State the type of reaction that will occur.

.....[1]

- (iii) Use the data given to write a chemical equation to show how the oxides of **Y** reacts with sodium hydroxide.

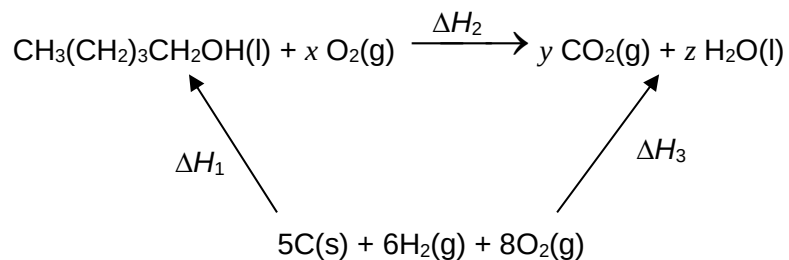
.....[1]

- (iv) Draw 2 simple diagrams to show how a water molecule can be attached to a sodium cation, and to an oxide anion. Label each diagram to show the type of interaction involved.

[2]

[Total: 14]

- 2 (a) Consider the following energy cycle involving pentan-1-ol,  $\text{CH}_3(\text{CH}_2)_3\text{CH}_2\text{OH}$ .



- (i) Define the term *standard enthalpy change of combustion* of pentan-1-ol.

.....  
 .....  
 .....[1]

- (ii) State the values of  $x$ ,  $y$  and  $z$ .

$x$ : .....  $y$ : .....  $z$ : ..... [1]

- (iii) What does  $\Delta H_1$  represent?

.....[1]

- (iv) Using the following data, calculate  $\Delta H_1$ .

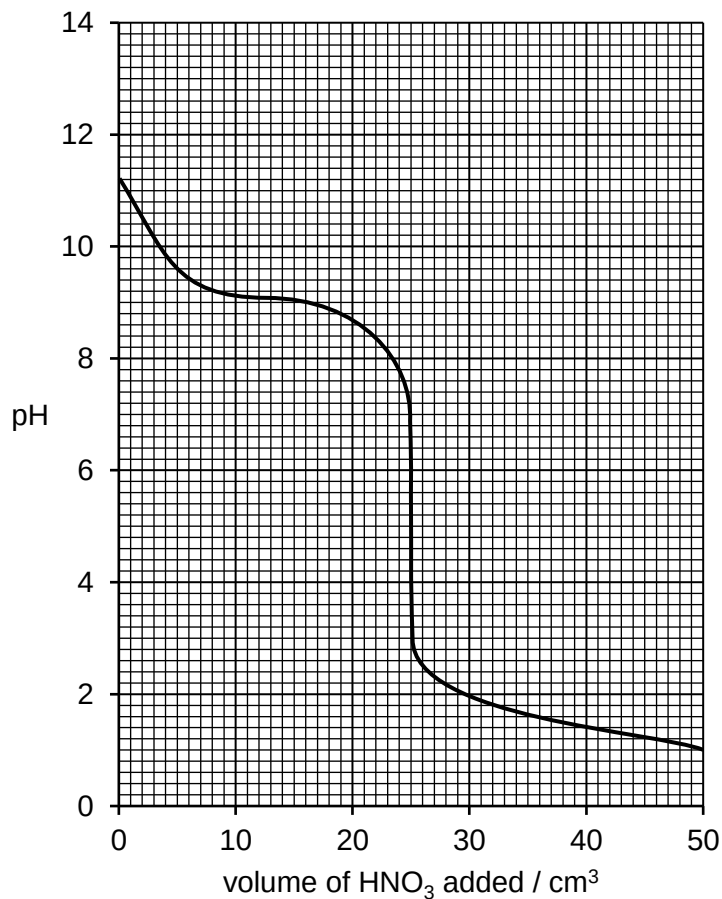
|                                              |                              |
|----------------------------------------------|------------------------------|
| enthalpy change of combustion of carbon      | = -393 kJ mol <sup>-1</sup>  |
| enthalpy change of combustion of hydrogen    | = -285 kJ mol <sup>-1</sup>  |
| enthalpy change of combustion of pentan-1-ol | = -3331 kJ mol <sup>-1</sup> |

[2]

- (b) Ammonia,  $\text{NH}_3$  is commonly used as a household cleaner to remove greasy dirt in ovens. It is also used to clean windows as it evaporates quickly and does not leave stains on glass.

$25.0 \text{ cm}^3$  of  $0.500 \text{ mol dm}^{-3} \text{ NH}_3$  was titrated against  $0.500 \text{ mol dm}^{-3} \text{ HNO}_3$ .

Fig. 2.1 shows the pH changes during the titration.



**Fig. 2.1**

- (i) Using the pH from Fig 2.1, calculate the concentration of hydroxide ions, in  $\text{mol dm}^{-3}$ , in  $25.0 \text{ cm}^3$  of  $0.500 \text{ mol dm}^{-3}$  solution of  $\text{NH}_3$ .

[1]

- (ii) Explain how your answer in (b)(i) indicates that  $\text{NH}_3$  is a weak base.

.....  
 .....[1]

- (iii) Using Fig. 2.1, state the volume of nitric acid required to reach the equivalence point for this titration.  
 Hence, suggest the concentration of the nitric acid use in the titration.

Volume of nitric acid required: .....

Concentration of nitric acid: ..... [1]

- (iv) Table 2.1 shows some acid–base indicators.

**Table 2.1**

| indicator       | colour in acid | colour in alkali | pH range of colour change of indicator |
|-----------------|----------------|------------------|----------------------------------------|
| alizarin yellow | yellow         | orange           | 10.1 – 13.0                            |
| methyl orange   | red            | yellow           | 3.1 – 4.4                              |
| phenolphthalein | colourless     | pink             | 8.2 – 10.0                             |

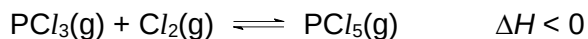
Suggest which of the above indicator should be used for this titration.  
 State the colour change of your chosen indicator at the end–point of this titration.

indicator .....

colour change ..... [2]

[Total: 10]

- 3 (a) Vapour of 0.2 mol phosphorus(III) chloride, 0.15 mol chlorine and 0.3 mol phosphorus(V) chloride can exist in equilibrium in a 5.0 dm<sup>3</sup> vessel at 230 °C.



- (i) Calculate the value of  $K_c$  at 230 °C, stating its unit.

[2]

- (ii) Suggest, with reasoning, whether lowering the temperature would favour the formation of  $\text{PCl}_5$ .

.....  
 .....  
 .....[1]

- (b) Phosphorus(V) chloride dissolves in polar solvents forming two ions  $[\text{PCl}_4]^+$  and  $[\text{PCl}_6]^-$ . Draw the dot-and-cross diagram and shape of  $[\text{PCl}_4]^+$ .

| Ion                | Dot-and-cross diagram | Diagram of shape |
|--------------------|-----------------------|------------------|
| $[\text{PCl}_4]^+$ |                       |                  |

[2]

[Total: 5]



**Question 4 starts on the next page**

- 4 (a) Carbon particles, oxides of carbon and nitrogen are commonly produced in the internal combustion engines of cars. These substances have significant environmental consequences if emitted into the atmosphere. Hence, catalytic converter is fitted to the exhaust to reduce the amount of polluting substances emitted.

Fig. 4.1 shows a catalytic converter that include a particle filter to remove solid particles of carbon.

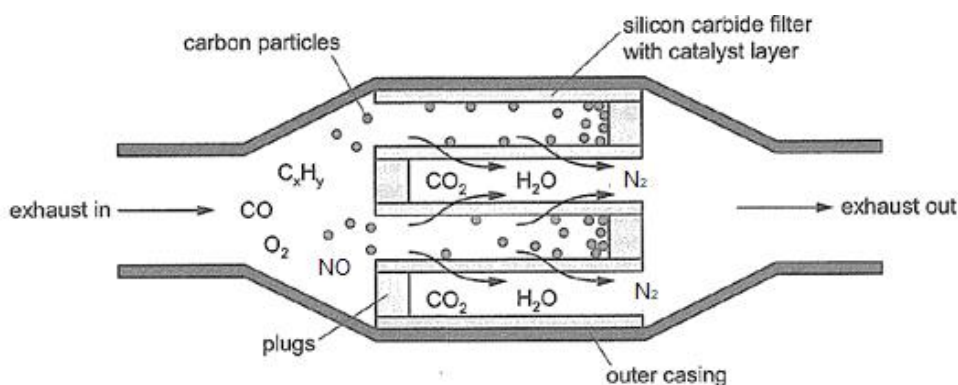
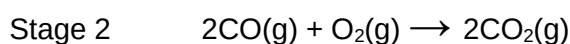
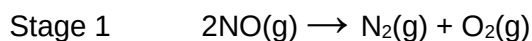


Fig. 4.1

- (i) The catalytic converter can remove nitrogen monoxide from the exhaust gases along with carbon monoxide, CO, which is also formed.

The conversion of NO and CO into  $N_2$  and  $CO_2$ , which occurs in two stages, has an overall  $\Delta H_{\text{reaction}} = -225 \text{ kJ mol}^{-1}$ . The oxygen produced in stage 1 is used in stage 2. Both stages are exothermic.



Complete the reaction pathway diagram on Fig. 4.2 for the overall reaction between NO and CO which involves both of these stages.

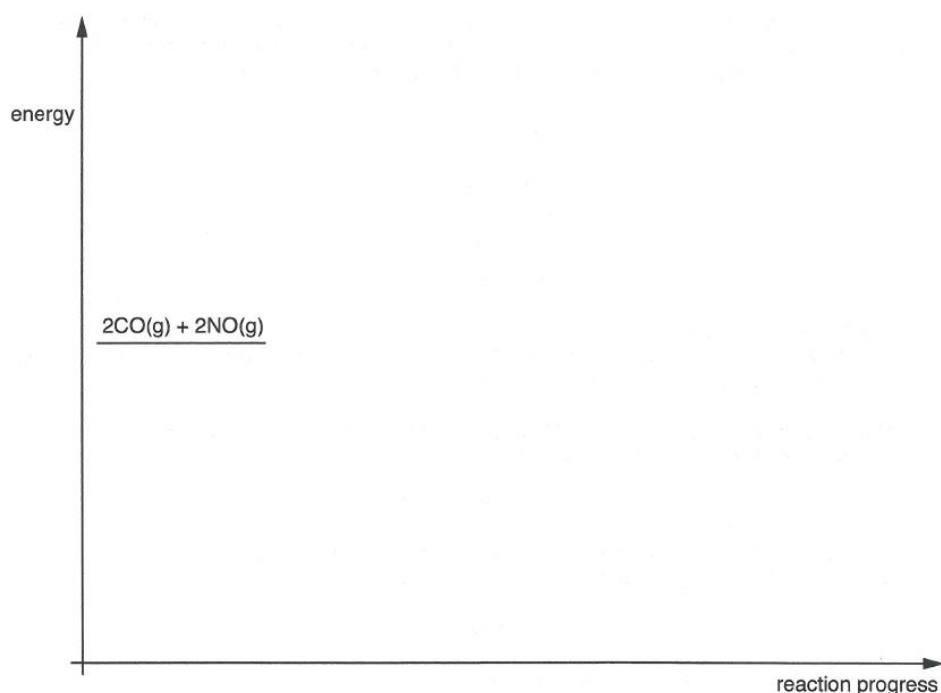


Fig 4.2

[2]

- (ii) Write an equation to show how unburnt hydrocarbon,  $C_xH_y$  can be removed.

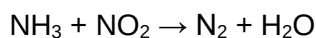
.....[1]

- (iii) Suggest why nitrogen monoxide is produced when petrol is burned in the internal combustion engine but not produced when petrol is burned in a spirit burner in a laboratory.

.....

.....[1]

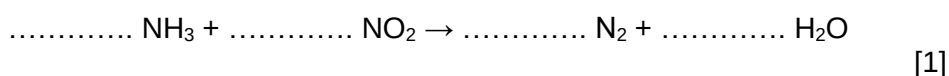
- (b) Diesel engines may also be fitted with a second catalytic converter to remove the remaining oxides of nitrogen. The second converter uses ammonia as the reducing agent. One possible reaction is shown by the following **unbalanced** equation.



- (i) Deduce all the changes in oxidation number, stating which element is involved in the change.

.....  
 .....[1]

- (ii) Hence, balance the equation.



- (c) Silicon carbide, SiC, is strong and resistant to melting at high temperature of the exhaust. Hence, it is a suitable material use for making the filter and for coating the Pt catalyst.

- (i) Suggest how the structure and bonding in SiC gives its properties above.

.....  
 .....  
 .....[2]

- (ii) Clusters of platinum catalyst coated onto fine powder of SiC in the converter is found to be around 4 nm in size.

Explain how this greatly increase the catalytic activity of the Pt catalyst.

.....  
 .....  
 .....[2]

[Total:10]

**Question 5 starts on the next page.**

- 5 (a) Infra-red spectroscopy is a fast, reliable and cost-effective technique suitable for analysis of polymers. The infra-red absorption frequencies in the spectrum give information about the presence or absence of specific functional groups in a molecule and can be used to determine the identity of the sample.

Three samples of polymers namely poly(vinyl alcohol) (PVA), polyethylene terephthalate (PET) and polyacrylamide (PAM) were taken and analysed using infra-red spectroscopy.

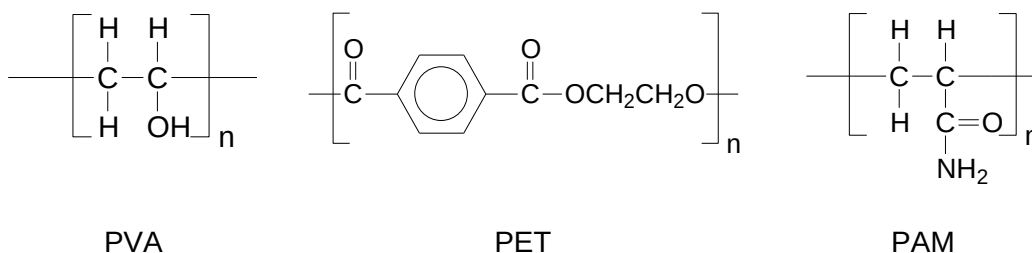


Table 5.1 shows some infra-red absorption frequencies in the spectra obtained from the three polymers.

**Table 5.1**

| Spectrum | Infra-red absorption frequencies / $\text{cm}^{-1}$ |
|----------|-----------------------------------------------------|
| <b>A</b> | 3100, 1740, 1050                                    |
| <b>B</b> | 3300, 1690                                          |
| <b>C</b> | 3200, 1100                                          |

Using relevant data from the *Data Booklet*, assign the identity of the three polymers to the relevant spectrum. Explain your answer.

| Spectrum | Polymer | Explanation |
|----------|---------|-------------|
| <b>A</b> |         |             |
| <b>B</b> |         |             |
| <b>C</b> |         |             |

[6]

- (b) PVA can be cross-linked with pentane-1,5-dial to form a three-dimensional network of polymer known as hydrogel. The hydrogel can swell in water and hold a large amount of water while maintaining the structure. Water is a large constituents of human body. Hence, hydrogel can be used for drug delivery in pharmaceutical field.
- (i) Draw a simple well-labelled diagram to present the cross-links in hydrogel made from PVA and pentane-1,5-dial.

[Structural formulae of molecules are **not necessary** in your diagram.]

[2]

.....[2]

.....[1]

Type of polymerisation : .....

| Monomer 1 | Monomer 2 |
|-----------|-----------|
|           |           |

---

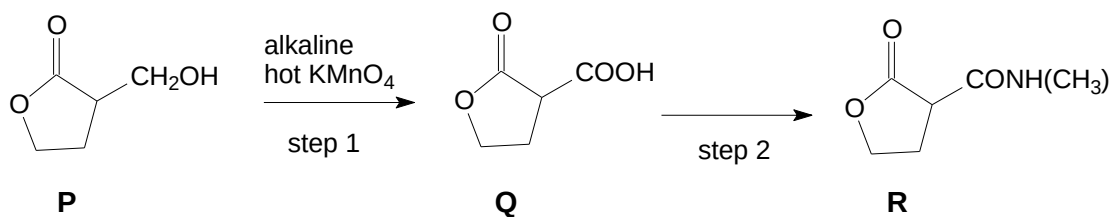
[3]

.....[1]

.....[1]



- 6 A student proposed the following reaction scheme to synthesise **R** from **P**.



- (a) State the types of reaction in steps 1 and 2.

Step 1 : .....

step 2 : .....

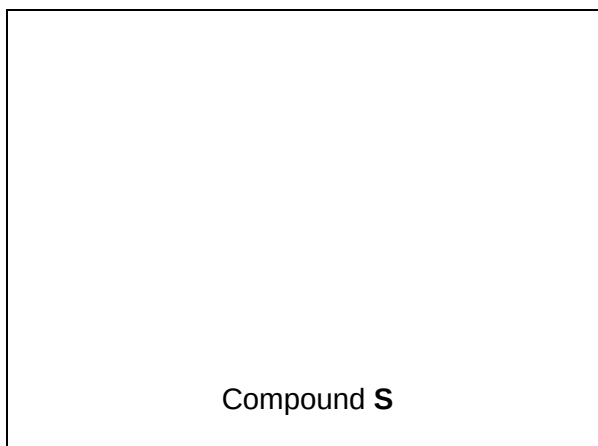
[2]

- (b) Suggest the reagents and conditions for step 2.

.....[1]

- (c) The student found that compound **Q** was not obtained as a major product in step 1 as shown in the reaction scheme. Instead, compound **S** was obtained.

Suggest the structure of compound **S**.



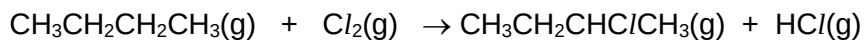
[2]

[Total: 5]

## Section B

Answer **one** question from this section, in the spaces provided.

- 7 (a) When briefly exposed to a bright light, a mixture of butane gas and chlorine gas reacts to give 2-chlorobutane and hydrogen chloride gas.

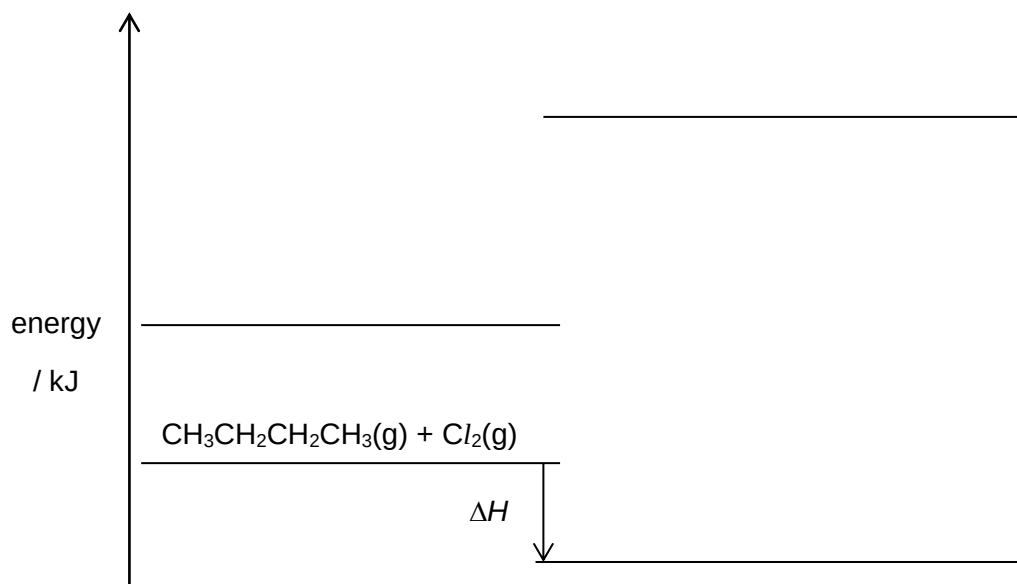


- (i) Use relevant bond energy values given in the *Data Booklet* to calculate the enthalpy change for this reaction,  $\Delta H$ .

[1]

- (ii) Draw arrows on the energy diagram below to show each of the bond breaking and bond making steps you have used in your calculation above.

Label each level with the appropriate formulae.



[3]

- (iii) Suggest why the reaction only starts when it is exposed to bright light.

.....  
 .....[1]

- (iv) State the type of reaction between butane and chlorine.

type of reaction : .....[1]

- (v) Suggest why the reaction continues with **increasing rate** after the **brief** exposure to the bright light has stopped.

.....  
 .....[1]

- (vi) In the reaction of butane with chlorine, another mono-chlorinated compound **S** is formed. Draw the structural formulae of this mono-chlorinated compound **S**.

Hence, suggest the statistical ratio of this compound **S** and 2-chlorobutane.

ratio of compound **S** : 2-chlorobutane = ..... : .....

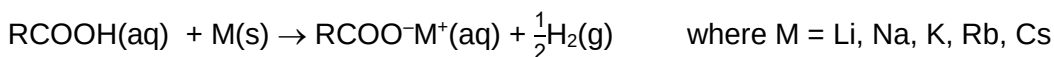
[2]

- (b) Under suitable conditions, 2-chlorobutane can be converted into two structural isomers of butene, of which one exhibit cis-trans isomerism.
- Suggest the reagent and condition for this reaction.
  - Draw the displayed formula of the pair of cis-trans isomers.

reagent and conditions: .....

[3]

- (c) Carboxylic acid reacts with Group 1 metals in the following reaction.



- (i) Suggest how you would expect the reactivity of Group 1 metals with carboxylic acid to change down the group. Explain your answer in terms of its reducing ability.

.....

.....

.....

.....

.....

.....

.....

.....

.....[3]

- (ii) Suggest with reasoning, whether you expect the same volume of hydrogen gas to be evolved in each case when excess carboxylic acid of same concentration is added separately to 5 g of different Group 1 metal powder.

[No calculation is required.]

.....

.....

.....

.....

.....

.....[3]

- (iii) Group 1 metals are often used in borohydrides,  $\text{LiBH}_4$ ,  $\text{NaBH}_4$  and  $\text{KBH}_4$ , as reducing agents in organic chemistry.

The greater the charge density of the metal cation, the stronger the reducing power of the borohydrides.

Identify the strongest reducing agent among the three Group 1 borohydrides listed above and explain your answer.

.....

.....

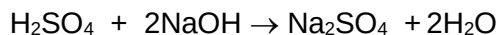
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.....

.....[2]

[Total: 20]

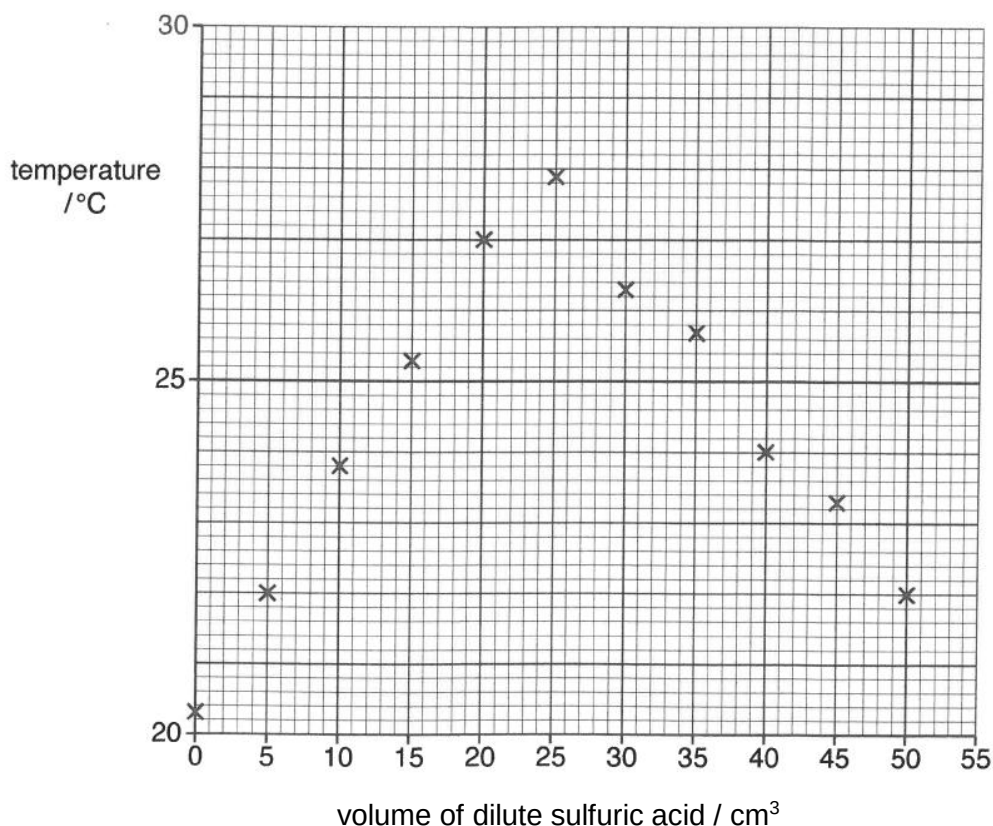
- 8 (a) In an experiment to determine the maximum change in temperature when sulfuric acid is added to sodium hydroxide, 30 cm<sup>3</sup> of sodium hydroxide is transferred to a polystyrene cup.



The initial temperature of the sodium hydroxide before sulfuric acid is added is measured and recorded.

5.0 cm<sup>3</sup> portions of 0.7 mol dm<sup>-3</sup> dilute sulfuric acid are added to the polystyrene cup. The temperature of the solution in the cup is measured after each addition. The maximum temperature occurs when complete neutralisation has been achieved.

The results are shown in Fig. 8.1



**Fig. 8.1**

- (i) Draw two separate straight lines of best fit on Fig. 8.1.  
The first line should take into account the increasing temperatures.  
The second line should take into account the decreasing temperatures.

Extend these lines until they intersect.

Use your graph to determine the maximum temperature rise,  $\Delta T_{\text{max}}$ , of the experiment. Show your working.

$$\Delta T_{\text{max}} = \dots\dots\dots^{\circ}\text{C} [2]$$

- (ii) Use your graph to determine the volume of sulfuric acid required to neutralise the sodium hydroxide and hence, calculate the concentration of sodium hydroxide.

Show on your graph how you determined this volume.

volume of sulfuric acid = .....  $\text{cm}^3$

concentration of sodium hydroxide = .....  $\text{mol dm}^{-3}$   
[3]

- (iii) Using your answers to (a)(i) and (a)(ii), calculate the energy, in kJ, evolved during the reaction.

You should assume that the specific heat capacity of the reaction mixture is  $4.18 \text{ J g}^{-1} \text{ K}^{-1}$  and the density of each solution is  $1.00 \text{ g cm}^{-3}$ .

[1]

- (iv) Using your answer to (a)(iii), calculate a value for the *enthalpy change of neutralisation*, in  $\text{kJ mol}^{-1}$ , for the reaction between sulfuric acid and sodium hydroxide.

[1]

- (v) Suggest how the student's calculated value for *enthalpy change of neutralisation* in (a)(iv) differ from the literature value.

.....  
.....  
.....[1]

- (b) When aqueous chlorine is added to an acidic solution containing  $\text{Mn}^{2+}$  ions, a black precipitate of  $\text{MnO}_2$  is formed.

- (i) Write an equation for the reaction between  $\text{Mn}^{2+}$  ions and aqueous chlorine.

.....[2]

- (ii) Explain why this is a redox reaction, in terms of oxidation numbers.

.....  
.....[1]

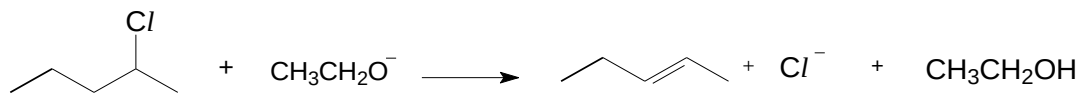
- (iii) There is no redox reaction when bromine is added to a solution containing  $\text{Mn}^{2+}$  ions.

Explain the difference in observations between chlorine and bromine with  $\text{Mn}^{2+}$  in terms of their ease in gaining of electrons.

.....  
.....  
.....  
.....  
.....  
.....[3]



- (c) 2-chloropentane reacts with  $\text{CH}_3\text{CH}_2\text{O}^-\text{Na}^+$  in the solvent ethanol according to the following equation.



- (i) Name the type of reaction that occurred.

Type of reaction: .....[1]

- (ii) Draw the two isomers of the product pent-2-ene and name the type of isomerism they exhibit.

Explain, with reference to one of your diagrams, why pent-2-ene show the type of isomerism.

|  |  |
|--|--|
|  |  |
|--|--|

Type of isomerism: .....

.....

.....

.....

.....

..... [4]

- (iii) State the observation when pent-2-ene reacts with aqueous bromine.

Observation: .....

.....[1]

[Total: 20]

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